Lab 2

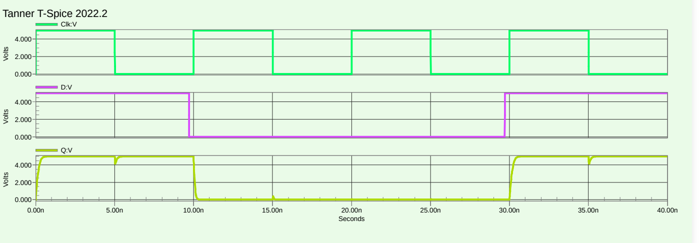
1

a

Is the circuit in Figure P1.2 positive or negative edge-triggered DFF? Describe its functionality using appropriate waveforms on nodes D, Q, and CLK (specify the clock frequency).

✓ Answer

It is positive edge triggered as the internal node holds the input voltage when the clock is low, but only updates the output once the clock turns high.



b

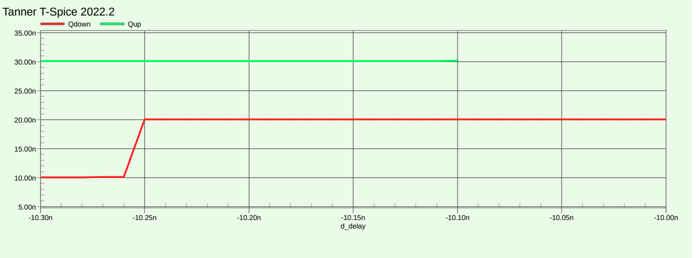
Determine the setup time T_{setup} and T_{hold} for the DFF; Consider both low-to-high and high-to-low data transitions.

✓ Answer

$T_{setup} = 0.10\ ns$ for the upward edge
$T_{setup} = 0.26\ ns$ for the downward edge

Both of the hold times are the opposites of each other, being negative they do not matter.

I measured this by performing a transient sweep on the offset of the data signal in reference to the clock signal. The graph below shows the transition times of the output Q given a D offset. At -10ns, D would line up exactly with the clock, which is why I began sampling from -10.30ns. Where you see the changes in the graph as the delay decreases is where Q fails to flip in regard to D, thus pointing us to the setup time required.



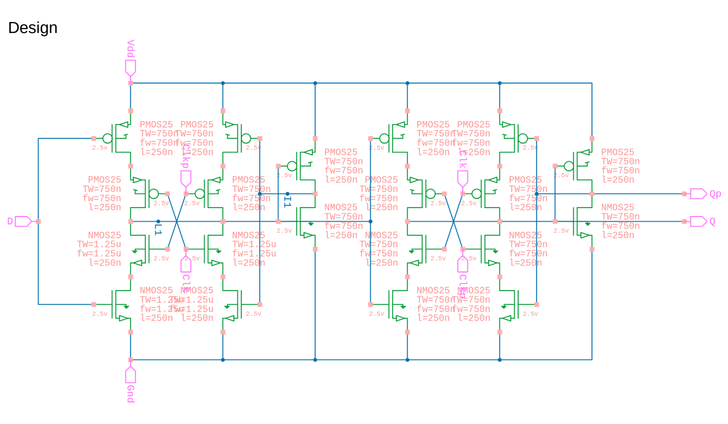
c

Measure the propagation delay for from clk to Q

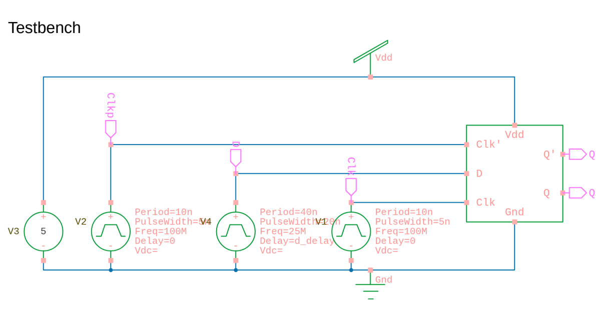
✓ Answer

Using a cursor, I am able to see that Q switches at 10.063ns, when the clock was triggered at 10ns, making a propagation delay of 0.063ns for the downward edge of Q. For the upward edge, we have a delay of 0.116ns using a similar method.

Design

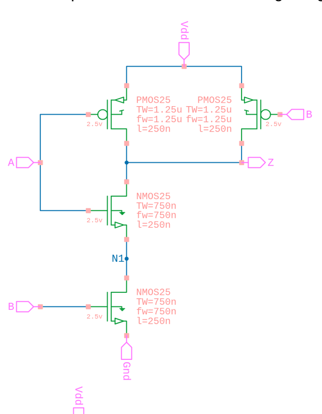


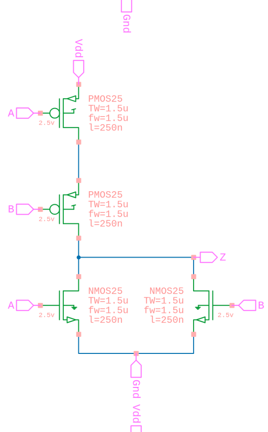
Testbench

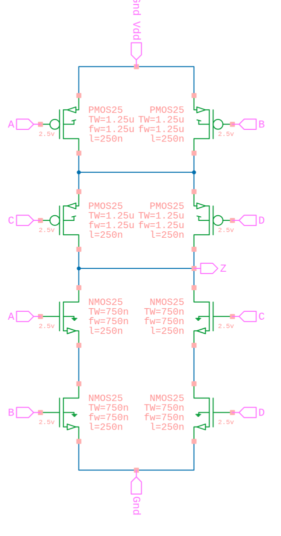


2

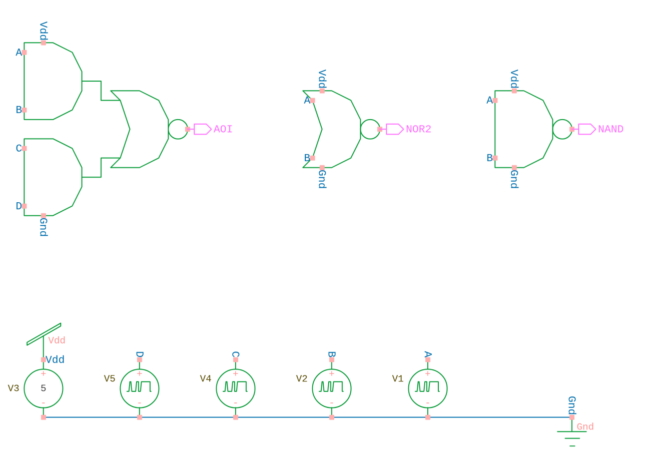
For this question we used the following designs:



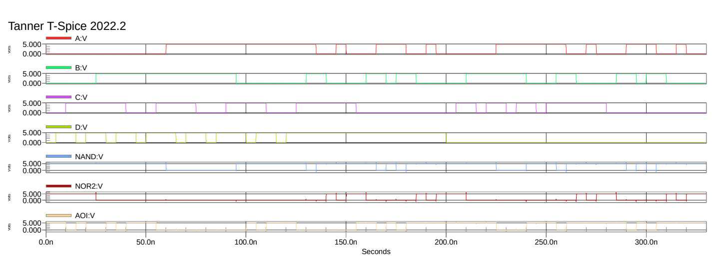




Testbench:



Waveform:



a-b

The best and worst case for t_{rise}, t_{fall}, and t_d.

✓ Answer

For NAND:

	Delay	Rise	Fall
Max	0.0435	0.0696	0.0513
at	10-11	11-10	10-11
Min	0.0256	0.0441	0.0505
at	11-10 11-01	11-01	01-11

For NOR2:

	Delay	Rise	Fall
Max	0.0692	0.1108	0.0389
at	10-00	10-00	00-10
Min	0.0178	0.1102	0.0219
at	00-01	01-00	00-01

For AOI:

	Delay	Rise	Fall
Max	0.1264	0.2206	0.1622
at	1110-1010	1110-1010	0110-1110
Min	0.0385	0.0891	0.0622
at	0001-0011	0011-0001	0001-0011

c

Explain your reasoning for a and b

✓ Answer

For the NAND, it makes sense that the maximal delay is 10-11 as it activates the furthest gate from the output, making the most delay before the output switches, the same logic makes sense for it being the slowest fall time. For the rise time, its equivalent if we went from 11-01 or 11-10, but experimentally we found 11-01 to be slower.

The same but opposite logic could be used for the NOR2 gate as it is functionally the same. We have different numbers for this as the p and n gates are different sizes and saturate slightly differently.

For the AOI, since we already know the p gates are slightly slower than the n gates, it would be slowest if we were rising our circuit, leading to either A or B activating Z overall as they are they furthest from Z. Experimentally we found B to be slower though.

Data Tables

NAND

Delay	Rise	Fall
0.0335		0.0506
0.0374	0.0692	
0.0435		0.0511
0.0256	0.0441	
0.0335		0.0505
0.0374	0.0691	
0.0434		0.0513
0.0256	0.0445	
0.0336		0.0505
0.0374	0.0696	
0.0435		0.0512
0.0256	0.0448	
0.0336		0.0505
0.0374	0.0691	
0.0433		0.0511
0.0256	0.0444	

NOR2

Delay	Rise	Fall
0.0179		0.022
0.0526	0.1102	
0.0211		0.0389
0.0692	0.1108	
0.0178		0.0219
0.0525	0.1102	
0.0211		0.0389
0.0689	0.1108	
0.0178		0.022
0.0525	0.1102	
0.0211		0.0389
0.0689	0.1108	
0.0178		0.022
0.0525	0.1104	
0.0211		0.0389
0.0689	0.1108	

AOI

Delay	Rise	Fall
0.0385		0.0622
0.0562	0.1215	
0.0484		0.0627
0.0677	0.1746	
0.0484		0.0624
0.0447	0.1316	
0.0386		0.0624
0.091	0.2151	
0.0544		0.0916
0.0677	0.1745	
0.0446		0.0918
0.0391	0.0891	
0.0626		0.1325
0.1031	0.1815	
0.0726		0.1312
0.0802	0.1418	
0.0743		0.1622
0.1264	0.2206	
0.0842		0.1605
0.103	0.1816	
0.0763		0.1362
0.0843	0.1655	
0.0862		0.1346
0.0669	0.1352	

3

a

Using the circuit in Figure 3.1 and the measured t_{rise}, t_{fall}, and t_d find the critical path.

✓ Answer

The longest path would be from IH(GDE)CA from Out to I0.

Our new t_d would be 0.2391 ns. (sum of all delays).

For calculating the rist and fall time of our output we typically can take the square root of the squares of all components on the way to the output (euclidian distance).

This gives us the following values:

t_{rise} of 0.2565 ns
t_{fall} of 0.1745 ns

b

Using the critical path and the set-up/hold time compute the clock cycle time.

✓ Answer

To calculate the total time, we need to sum our: total combinational delay and register hold time or our register setup time, since one runs during the upward cycle and one during the downward cycle.

Since our register hold time is negative, we do not need to account for that.

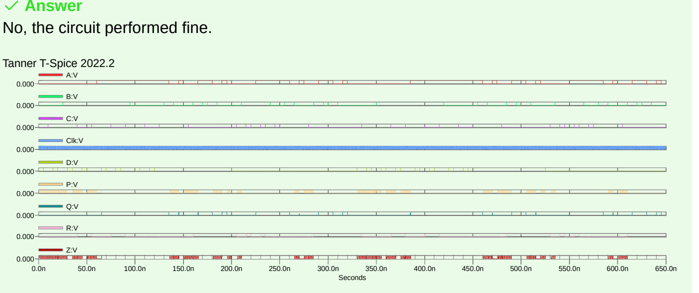
Our total critical path should be 0.2391 ns for the upward cycle and 0.26ns for the downward cycle, which leaves us with a clock of 2.00GHz (asymmetric square wave). Assuming a symmetrical clock, we would be limited to 1.92 GHz.

c

Using the computed clock cycle time, does the circuit has any set-up or hold-time violation.

✓ Answer

No, the circuit performed fine.



d

Briefly explain in one paragraph how you found the critical path and explicitly give the worst case path delay.

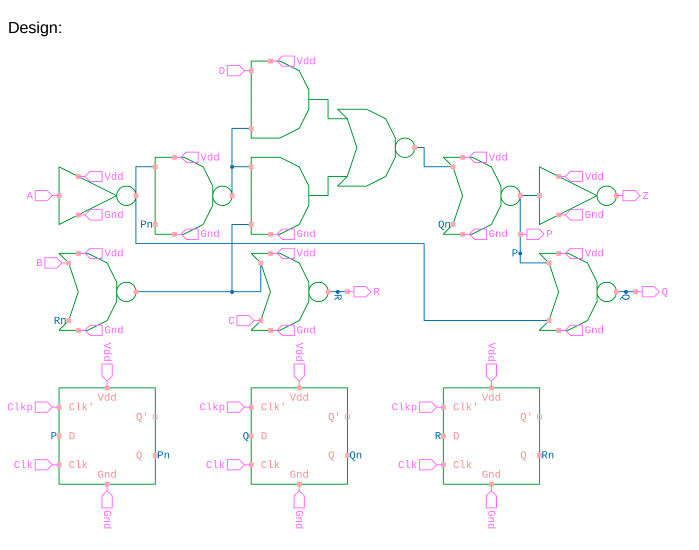
✓ Answer

Explained prior, we looked at the delays, rise, and fall times of the components within our circuit, then calculated the maximal path from start to finish. The worst case would be:

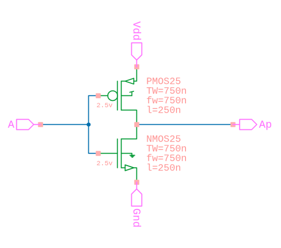
A: 0->1
B: 1
C: X
D: 1

P: 1
Q: 0
R: X

Design:



Inverter:



Testbench: